

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Borehole Geophysics L-T-P: 3-0-0

Credits:3

3. Pre-requisites: NA

4. Syllabus and reference books:

Syllabus:

Fundamental concepts in borehole geophysics – formation factor, porosity, permeability, formation water resistivity, water saturation, irreducible water saturation, hydrocarbon saturation, residual hydrocarbon saturation, Archie's and Humble's equations. Principles, techniques and interpretation of SP, Resistivity and Microresistivity logs, Nuclear logs, Acoustic impedance, Temperature log, Caliper log, Dip and Directional logs. Evaluation of different parameters, Clean sand and Shaly sand interpretation. Applications in groundwater, metallic and non-metallic exploration. Computation of resistivity log responses. Estimation of porosity, permeability, formation water resistivity, water saturation, hydrocarbon saturation, moveable oil, Gas corrections. Interpretation of log.

Lithodensity log, neutron activation technique, thermal neutron decay time log, pulse neutron log, compensated neutron log. Recording, transmission and processing of log data. Formation evaluation for hydrocarbons. Qualitative and quantitative interpretations of well log data: modelling of shaly formations, Indonesian equations and other popular nonlinear models. Overlays and cross-plots. Determination of reservoir parameters - porosity, resistivity, permeability, water and hydrocarbon saturation, movable oil. Lithology determination by neutron, density and sonic cross-plots, dual mineral method, triporosity method, lithoporosity cross-plot (M-N plot), Clean sand and shaly sand interpretations. Sub-surface correlation and mapping from log data. FMI Logging, Acoustic Imaging and Electrical imaging logs: thin bed detections, blow out identifications and delineation of fractures from logs. NMR Logging Production logging. Cement Bond Logging. LWD. Well logging for non- hydrocarbon Geophysical applications. Shale gas interpretations using well logs. API units, API numbering, TVT, TVDT. Carbonates logging and interpretations techniques.

Reference Books:

- 1) Fundamentals of Well-log interpretation 1st Edition by O Serra, ELSEVIER (1984)
- 2) Geophysics Under Stress: Geomechanical Applications of Seismic and Borehole Acoustic Waves, 1st Edition by Colin M Sayers, SEG Vol-13 (2010)

3) Principles and Applications of Well Logging 2nd Edition by LIU, SPRINGER (2017)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Fundamental concepts in borehole geophysics	3
2.	Principles, techniques and interpretation of SP, Resistivity and Microresistivity logs, Nuclear logs, Acoustic impedance, Temperature log, Caliper log, Dip and Directional logs.	6
3.	Evaluation of different parameters, Clean sand and Shaly sand interpretation.	3
4.	Applications in groundwater, metallic and non-metallic exploration	2
5	Computation of resistivity log responses.	2
6	Estimation of porosity, permeability, formation water resistivity, water saturation, hydrocarbon saturation, moveable oil, Gas corrections. Interpretation of log.	6
7	Lithodensity log, neutron activation technique, thermal neutron decay time log, pulse neutron log, compensated neutron log.	6
8	Formation evaluation for hydrocarbons. Qualitative and quantitative interpretations of well log data: modelling of shaly formations, Indonesian equations and other popular nonlinear models.	3
9	Overlays and cross-plots. Determination of reservoir parameters - porosity, resistivity, permeability, water and hydrocarbon saturation, movable oil.	3
10	Lithology determination by neutron, density and sonic cross-plots, dual mineral method, triporosity method, lithoporosity cross-plot (M-N plot), Clean sand and shaly sand interpretations. Sub-surface correlation and mapping from log data.	3
11	FMI Logging, Acoustic Imaging and Electrical imaging logs: thin bed detections, blow out identifications and delineation of fractures from logs.	3
12	NMR Logging Production logging. Cement Bond Logging. LWD.	3

13	Well logging for non- hydrocarbon Geophysical applications. Shale gas interpretations using well logs. API units, API numbering, TVT, TVDT.	3
14	Carbonates logging and interpretations techniques.	2
Total number of hours		48